

AYMANE AIT DADS

AI / Gen AI Engineer Intern | LLM Fine-Tuning · Graph-Contextual AI · Agentic Pipelines

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PROFESSIONAL SUMMARY

Ranked **17th out of 3,000+ teams** in the live NVIDIA Nemotron Reasoning Challenge (score 0.86/1.0, \$106K+ prize pool) by engineering an end-to-end LoRA SFT pipeline on a 30B-parameter Mamba-Transformer - directly demonstrating the LLM training and contextual reasoning depth Arango requires. Reduced manual network analysis time by **60%** at Orange Maroc through production-grade ML pipelines on 100K+ records, translating model outputs into actionable business recommendations for enterprise stakeholders. Brings hands-on expertise in agentic pipelines, RAG, LangChain, Claude API tool-calling, and graph-aware LLM orchestration - skills that map directly to Arango's mission of building trusted, context-rich data foundations for enterprise AI. Aims to contribute to Arango's Contextual AI platform by applying verified chain-of-thought dataset engineering, ablation-driven model evaluation, and enterprise-scale LLM deployment experience.

CORE COMPETENCIES

Contextual AI | LLM Fine-Tuning | Agentic Pipelines | RAG | Graph-Based AI | Enterprise AI | Prompt Engineering | LLM Orchestration

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Engineered a **production-grade ML pipeline** combining Random Forest and K-Means on 100K+ fiber-optic network records, classifying quality across thousands of nodes for the network optimization team.
- Developed a clustering pipeline that mapped customer performance profiles to **5 commercial service tiers**, with outputs adopted directly by stakeholders for infrastructure decisions.
- Reduced manual analysis time by **~60%** through automated feature engineering, model evaluation workflows, and reusable Pandas/Scikit-learn data processing modules.
- Delivered stakeholder presentations translating complex model outputs into **enterprise maintenance recommendations**, bridging data science findings and business decision-making.

PROJECTS

NVIDIA Nemotron Model Reasoning Challenge (In Progress)

Kaggle · Live Competition June 2026 · \$106K+ Prize Pool

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, \$106K+ prize pool) fine-tuning a 30B-parameter Mamba-Transformer with LoRA SFT ($r=32$, BF16, completion-only loss, 8,192-token context) on 5 structured reasoning categories.
- Ran **controlled ablation experiments** across LR schedules, packing strategies, and synthetic augmentation - discovering reasoning-style consistency outweighs data volume in LLM SFT; built custom cipher parsers and 8-bit operation solvers to produce verified chain-of-thought training data.

PyTorch · Unsloth · TRL · PEFT · Hugging Face Transformers · vLLM · Triton

AI-Generated Image Detection - NTIRE 2026 @ CVPR

EURECOM ImSecu + NTIRE 2026 CodaBench · 2025 · Ranked 1st in Class

- Ranked **1st in the EURECOM class private Kaggle leaderboard** (0.791 AUC) fine-tuning CLIP ViT-L/14 (428M parameters) with a custom MLP head and dual learning-rate strategy ($2e-6$ backbone, $5e-4$ head) on 250K images spanning 25+ generator types.
- Submitted independently to the **NTIRE 2026 @ CVPR CodaBench** international benchmark; achieved best generalization at 1 epoch (0.793 AUC vs. 0.767 AUC at 4 epochs), with a 3-model ensemble and 10-view TTA improving out-of-distribution robustness.

PyTorch · OpenCLIP · ViT · FP16 Mixed Precision · AdamW

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, NLP, Computer Vision, Cloud Computing, Statistics, Image Security, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

ML Frameworks & LLM Engineering: PyTorch | Hugging Face Transformers | Unsloth | TRL | PEFT | LoRA / QLoRA | vLLM

Agentic AI & Orchestration: LangChain | RAG | Claude API | Prompt Engineering | Tool-Calling | Agentic Pipelines | MCP Server

Data Science & MLOps: Python | Scikit-learn | Pandas | SQL | Power BI | Docker | Git | BF16/FP16 Mixed Precision